Olonade Kelvin

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Education

University of Lagos — **Lagos, Nigeria**
B.Sc. Electrical & Electronics Engineering — CGPA: 4.18 / 5.00 — *Expected Oct 2028*

Technical Skills

HDL & Design: Verilog, SystemVerilog, RTL Design, FSM Design, Synthesizable IP, Parameterized Modules
Verification: Self-checking Testbenches, Functional Coverage, Constrained-Random, Assertions, Edge-Case Analysis
EDA & Flow: OpenLane, Yosys, Magic, KLayout, Icarus Verilog, GTKWave, Verilator, OpenROAD, Cocotb
Process & PDKs: SkyWater SKY130, Standard-Cell Synthesis, STA Basics, Place & Route, DRC, LVS
Embedded & Tools: PIC16F CLB, CIP, ARM Cortex-M, C, C++, Python, Git, Linux, MATLAB, Simulink

Silicon & Hardware Projects

Autonomous VGA Algorithmic Synthesizer ASIC — Verilog, SKY130, OpenLane

- Designed a tape-out-ready digital ASIC for Tiny Tapeout SKU26A on SkyWater 130 nm, generating autonomous VGA 640×480 visuals from on-chip fractal and LFSR engines with zero firmware and pure digital logic.
- Hand-optimized RTL for extreme area constraints (1 tile $\approx 160 \times 100$ μm^2); pipelined the pixel pipeline at 25.175 MHz and balanced critical paths through OpenLane synthesis and STA.
- Implemented a self-checking SystemVerilog testbench with functional coverage on sync timing, fractal correctness, and entropy quality before signoff.

Synthesizable UART TX/RX IP Core — Verilog, Cocotb, Verilator

- Designed a fully parameterized UART IP (baud rate, data bits, parity, stop bits) achieving 100% functional coverage via a self-checking testbench with constrained-random stimulus and a golden-model reference.
- Verified back-to-back transmission, framing and parity error injection, and metastability handling across asynchronous domains using two-flop synchronizers.

Hardware EEPROM Wear-Leveling Controller — Verilog, SKY130, Tiny Tapeout

- Designed and taped out a compact, attack-resistant wear-leveling controller implementing the Start-Gap algorithm (used in 3D XPoint) to extend EEPROM/flash lifetime; verified with Cocotb and synthesized on SkyWater 130 nm via Tiny Tapeout SKU26A.
- Incorporated a 2-round Feistel address scrambler with LFSR key to defeat targeted wear attacks, featuring 8-bit per-block wear counters, bad-block retirement, and in-band telemetry for max-min wear skew and total write count.
- Achieved successful GDS generation with full place-and-route and functional simulation sign-off, proving a silicon-ready IP core with hardware-level security and reliability.

Urban FloodGuard — Firmware-Free Flood Monitor — PIC16F CLB, CIP

- Implemented a deterministic flood-alert system using Configurable Logic Blocks and Core Independent Peripherals exclusively, eliminating MCU firmware and demonstrating mastery of combinational and sequential digital design on real silicon.
- Validated logic correctness on real PIC16F silicon via oscilloscope captures and PIC simulator waveforms, sweeping a function generator across slow-rise, surge, noise, and sensor-fault conditions to cover all critical edge cases.
- Achieved zero false positives across the full test suite while delivering sub-microsecond sensor-trigger-to-alert latency via the CLC propagation path, eliminating polling-loop delay and idle-firmware power draw of a firmware-based equivalent.

Experience

Embedded Systems Engineer — **May 2026 – Present**
Balancee Tech Solutions — *Ajah, Lagos*

- Led end-to-end design of a custom fuel-pump embedded controller, encompassing schematic capture, MCU firmware, sensor signal-chain design, and field-test verification.

Embedded Systems Intern — **Dec 2025 – Present**
AVZDAX — *Victoria Island, Lagos*

- Delivered an edge CV pipeline on Raspberry Pi 5 and Pi 4B, integrating a Hailo-8 AI accelerator for all inference workloads on a custom-trained proprietary model via hardware-software co-design.
- Eliminated pipeline stalls through systematic bottleneck profiling across camera capture, pre-processing, and accelerator hand-off stages, achieving stable throughput well above the 30 FPS real-time target on the integrated Hailo-8 plus Pi 5 platform.

Leadership & Certifications

Established the first IEEE Solid-State Circuits Society (SSCS) Student Chapter at the University of Lagos as Founding Chair, organizing RTL/ASIC workshops and building Nigeria's pipeline of silicon engineers.

Arm Developer Program Contributor — authored technical content on Cortex-M architectures for the Arm developer community.

Certifications: Zero to ASIC Course, ARM Cortex-M Processors Overview, Logic Circuit Design 101 (Microchip), Rapid Prototyping with Curiosity Nano, Advanced Embedded C, UC Irvine IoT & Embedded Systems, Modeling & Simulation with Simulink.